

BUTUAN, PhilippinesWMO: **987520**Lat: **8.9472N**Lon: **125.4823E**Elev: **46**StdP: **100.77**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.4	23.1	20.9	15.7	24.4	21.6	16.3	24.8	9.5	25.9	7.4	27.3	0.7	140	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.6	34.2	27.5	33.7	27.4	33.2	27.3	28.5	32.6	28.1	32.3	27.9	31.9	3.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.3	23.2	31.2	27.0	22.8	30.9	26.7	22.4	30.6	92.2	32.7	90.6	32.5	89.4	32.2	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.8	5.1	20.2	35.9	2.2	0.9	18.6	36.6	17.3	37.1	16.0	37.6	14.4	38.2		
(5)			19.8	29.4	1.8	0.7	18.4	29.9	17.4	30.4	16.3	30.8	15.0	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	26.6	26.9	27.6	28.4	29.1	28.7	28.5	28.6	28.5	28.4	27.7	27.5
	DBStd	1.26	1.25	1.17	1.09	1.03	0.91	0.99	0.93	0.91	0.86	0.84	1.04	1.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6589	516	474	544	551	592	562	573	578	556	571	531	541
	CDD18.3	3547	258	241	286	301	334	312	315	319	306	312	281	283
	CDH23.3	38253	2200	2202	2906	3432	4053	3601	3611	3691	3498	3478	2838	2744
	CDH26.7	14253	576	664	1007	1373	1763	1464	1431	1499	1390	1321	929	836
Wind	WSAvg	1.7	1.5	1.6	1.7	1.7	1.9	1.8	1.9	2.0	2.0	1.8	1.5	1.6
Precipitation	PrecAvg	2273	329	238	154	119	130	195	169	140	151	171	220	257
	PrecMax	3460	804	508	360	299	243	445	356	228	398	365	456	631
	PrecMin	1407	33	19	6	11	74	67	66	47	37	66	84	82
	PrecStd	470	180	137	83	77	42	86	70	50	72	66	109	135
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	32.8	33.6	34.5	34.9	34.6	34.2	34.2	34.1	34.0	33.3	33.2
		MCWB	26.7	26.7	26.6	27.4	27.5	28.0	27.6	27.3	27.7	27.3	27.4	27.2
	2%	DB	31.5	31.8	32.6	33.6	34.1	33.6	33.3	33.4	33.2	33.1	32.4	32.2
		MCWB	26.5	26.4	26.6	27.2	27.6	27.7	27.3	27.2	27.4	27.2	27.1	26.9
	5%	DB	30.7	31.1	31.8	32.8	33.4	32.9	32.6	32.7	32.6	32.4	31.7	31.4
		MCWB	26.3	26.2	26.5	27.0	27.5	27.5	27.1	27.0	27.2	27.1	26.9	26.8
	10%	DB	29.7	30.3	31.1	32.1	32.7	32.2	31.9	32.1	31.9	31.7	30.9	30.5
		MCWB	26.1	26.0	26.3	26.9	27.4	27.3	27.0	26.9	27.0	26.9	26.7	26.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.7	28.1	28.5	28.7	28.7	28.5	28.4	28.6	28.5	28.1	28.1
		MCDB	31.1	31.2	31.9	32.6	33.0	33.0	32.4	32.7	32.8	32.2	31.8	31.7
	2%	WB	27.1	27.1	27.5	28.0	28.2	28.2	27.9	27.8	28.0	27.9	27.6	27.5
		MCDB	30.3	30.6	31.2	32.2	32.6	32.5	32.2	32.0	32.2	31.7	31.2	31.0
	5%	WB	26.7	26.7	27.0	27.6	27.9	27.7	27.6	27.5	27.6	27.5	27.3	27.1
		MCDB	29.6	30.0	30.7	31.7	32.2	31.9	31.7	31.6	31.6	31.3	30.7	30.4
	10%	WB	26.2	26.3	26.6	27.1	27.6	27.5	27.2	27.1	27.2	27.2	27.0	26.7
		MCDB	29.0	29.5	30.1	31.2	31.7	31.5	31.1	31.2	31.1	30.8	30.2	29.8
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.5	7.0	7.4	7.6	7.3	7.2	7.3	7.3	7.1	6.6	6.3
		MCDBR	7.6	8.0	8.3	8.5	8.5	8.3	8.2	8.2	8.2	8.1	7.8	7.7
		MCWBR	3.6	3.7	3.8	3.7	3.6	3.6	3.6	3.5	3.5	3.4	3.6	3.5
	5% WB	MCDBR	6.8	7.2	7.6	8.0	7.9	7.9	7.7	7.8	7.9	7.6	7.3	7.1
		MCWBR	3.5	3.7	3.8	3.8	3.6	3.7	3.7	3.6	3.8	3.6	3.6	3.5
Clear-Sky Solar Irradiance	taub	0.438	0.426	0.420	0.420	0.420	0.412	0.417	0.442	0.438	0.444	0.421	0.429	
	taud	2.385	2.403	2.432	2.427	2.444	2.475	2.455	2.367	2.373	2.356	2.441	2.414	
	Ebn at Noon	874	896	903	890	872	869	868	858	873	868	884	874	
	Edh at Noon	121	122	120	120	115	110	113	125	126	127	114	116	
All-Sky Solar Radiation	RadAvg	3.67	4.39	5.07	5.49	5.39	5.00	4.89	5.11	5.05	4.78	4.18	3.86	
	RadStd	0.51	0.49	0.43	0.40	0.38	0.32	0.32	0.32	0.40	0.34	0.39	0.35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page